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Musculoskeletal Adaptations to 16 Weeks of Eccentric Progressive Resistance Training in Young Women

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Abstract. Schroeder, E.T., S.A. Hawkins, and S.V. Jaque. Musculoskeletal adaptations to 16 weeks of eccentric progressive resistance training in young women. *J. Strength Cond. Res. 18(2): 227–235. 2004.*—We investigated the musculoskeletal adaptations and efficacy of a whole-body eccentric progressive resistance-training (PRT) protocol in young women. Subjects ($n = 37$; mean age, 24.3) were randomly assigned to one of 3 groups: high-intensity eccentric PRT (HRT), low-intensity eccentric PRT (LRT), or control. Subjects performed 3 sets of 6 repetitions at 125% intensity or 3 sets of 10 repetitions at 75% intensity in the HRT and LRT groups, respectively, 2 times per week for 16 weeks. Strength was determined by the concentric 1-repetition maximum (1RM) standard. Bone mass and body composition were measured by dual-energy x-ray absorptiometry (DXA). Blood and urine samples were obtained for deoxypyridinoline, osteocalcin, creatine kinase, and creatinine. Data were analyzed by repeated-measures analysis of variance with post hoc comparisons. Strength increased 20–40% in both training groups. Lean body mass increased in the LRT (0.7 ± 0.6 kg) and HRT (0.9 ± 0.9 kg) groups. Bone mineral content increased (0.855 ± 0.958 g) in the LRT group only. Deoxypyridinoline decreased and osteocalcin increased in the HRT and LRT groups, respectively. These findings suggest that submaximal eccentric training is optimal for musculoskeletal adaptations and that the intensity of eccentric training influences the early patterns of bone adaptation.

Key Words. bone mass, bone markers, muscle damage, strength training

Introduction

Improving peak bone mass accretion in young women is an important step in the prevention of osteoporosis, since maximizing bone mass in early adulthood may reduce the risk of osteoporotic fracture in senescence. Standard resistance-training protocols enhance muscle (15) and bone (5, 20) in young women. These conventional strength-training techniques involve both a concentric and an eccentric component. However, maximal skeletal muscle eccentric contractions develop greater tension than maximal concentric or isometric contractions (18). Consequently, the greater loads associated with eccentric contraction may be necessary to maximize musculoskeletal adaptations if, in fact, overload stimulus is required for compensatory gains. Indeed, studies have demonstrated that eccentric muscle contraction results in greater muscle strength and hypertrophy (9–11) and bone mass (8) than does concentric muscle contraction. Therefore, standard training techniques, which do not emphasize eccentric contractions, may not optimize muscle and bone adaptations.

The intensity of an eccentric-training protocol may have important osteogenic influences. Although we previously reported that the load magnitude associated with maximal eccentric contraction is responsible for significant increases in bone mass in young women (8), the mechanism(s) by which eccentric contraction may contribute to bone adaptation remains unknown. The effects of standard resistance training on bone mass are most commonly attributed to increased rates of bone formation (2, 6, 13, 16) and perhaps transiently suppressed rates of bone resorption (6). Whether a short time course of eccentric training differentially affects bone formation and resorption is uncertain.

It is well established that eccentric contraction results in muscle stiffness, prolonged strength loss, elevated circulating muscle proteins, and morphologic changes in muscle architecture (3). However, the relative contribution of eccentric contraction-induced muscle damage and the associated factors (e.g., elevated cytokines, enzymes) as a stimulus for musculoskeletal adaptation remains to be elucidated. Regardless, eccentric muscle contraction appears to be an important component in maintaining and enhancing the musculoskeletal system.

Previous investigations using eccentric-training models have limited applicability, since most of these studies trained participants using single muscle groups on isokinetic dynamometers. Although their findings provide valuable knowledge that contributes to the understanding of eccentric-training adaptations, they cannot describe the utility of eccentric resistance training as it applies to the use of standard gym equipment. Many athletes and recreational strength trainers do not have access to the proper equipment to allow them to train using isokinetic dynamometers; however, they routinely train with free weights and strength-training machines. Therefore, the purpose of this study was to determine the musculoskeletal adaptations and efficacy of a whole-body, multiple-muscle group, eccentric progressive resistance-training (PRT) protocol using free weights and standard strength-training machines in young women. We hypothesized that women participating in high-intensity eccentric resistance training would have significantly greater increases in lean body mass and muscle strength and improved bone adaptations compared to low-intensity eccentric resistance training.

Methods

Experimental Approach to the Problem

Since improving musculoskeletal strength in young women may help prevent or delay the risk of osteoporotic frac-

TABLE 1. Baseline characteristics of study subjects.*†

	LRT (n = 14)	HRT (n = 14)	Control (n = 9)
Age (y)	24.4 ± 1.9	24.0 ± 1.4	24.4 ± 2.2
Height (cm)	164.4 ± 6.5	165.8 ± 5.5	161.4 ± 8.8
Total body mass (kg)‡	56.8 ± 7.2	55.7 ± 6.0	58.2 ± 13.5
Lean body mass (kg)‡	40.7 ± 4.2	40.2 ± 3.8	40.8 ± 7.5
Body fat (%)‡	24.3 ± 3.4	23.6 ± 3.9	25.1 ± 5.6

* LRT = low-intensity eccentric progressive resistance training; HRT = high-intensity eccentric progressive resistance training.
† Mean ± 1 *SD*.
‡ Measured by dual-energy x-ray absorptiometry.

ture in senescence, we designed a resistance-training intervention to test the hypothesis that high-intensity eccentric-only compared to low-intensity eccentric-only training would maximize musculoskeletal adaptations. Therefore, changes in lean body mass, muscular strength, bone mass, and biochemical markers of bone metabolism were assessed to compare the 2 treatment groups performing very different intensities of eccentric resistance training.

Subjects

The volunteer subjects were women aged 18–28 years. Subjects were recruited from the University of Southern California (USC) Health Sciences Campus. They completed a brief informational entry questionnaire and, if they met the inclusion criteria, were contacted by the study staff. All subjects were free from conditions that would limit their participation in high-intensity resistance training, such as musculoskeletal injuries, hypertension, coronary heart disease, chronic pulmonary disease, osteoporosis, osteoarthritis. Additionally, all subjects were free from conditions known to influence the growth and maintenance of bone, including diabetes, eating disorders, pregnancy, polycystic ovarian disease, hyper- and hypothyroidism, hypercortisolism, amenorrhea, and any other hormonal or metabolic disorders. Menstrual cycles had to be regular (eumenorrheic) and between 26 and 31 days in length. Subjects were accepted regardless of oral contraceptive (OC) status. Subjects were nulliparous, and those who had a recent series of radiographs (such as radiation therapy) were excluded. The subjects must not have participated in resistance training 6 months prior to study entry. Subjects provided written informed consent prior to entry into the study, which was approved by the Institutional Review Board of the USC Medical Center at Los Angeles.

Thirty-seven of the 41 subjects who signed the informed consent completed the study. All subjects in the exercise groups and 9 of 14 subjects in the control group completed pre- and postmeasures. Reasons for subject withdrawal included time commitment (n = 1), phobia of blood draws or needles (n = 3), and unrelated illness (n = 1). All 28 subjects participating in eccentric training completed 32 workouts. Additionally, no subjects experienced any training-related injuries, except for 1 subject in the high-intensity eccentric PRT (HRT) group who reported an unusually stiff and sore biceps tendon, which resolved with 3 days of rest. Serum and urine samples were obtained at study weeks 4, 8, and 12 in the low-intensity eccentric PRT (LRT) group from 11, 13, and 12 subjects, respectively. At study weeks 4, 8, and 12 in the HRT group, 13, 13, and 11 subjects, respectively, provided blood and urine samples. As indicated, not all 14 subjects in each group were able to attend their scheduled appointment to provide blood and urine samples. Furthermore, 10 subjects in the LRT group, 11 subjects in the HRT group, and 4 subjects in the control group were taking OCs. The remaining subjects were not on OCs before or during the study intervention. The study groups were generally comparable in baseline characteristics (Table 1).

Study Design

The subjects were randomly assigned to one of 3 groups: HRT (n = 14), LRT (n = 14), or control (n = 9). All subjects were asked to visit the laboratory on 3 occasions. The pre-entry visit included densitometry, strength testing, and blood and urine samples. This visit was scheduled during the first 10 days of the subject's menstrual cycle. The second visit was a baseline visit to perform an additional strength test at least 5 days after the initial strength test. Lastly, all subjects were asked to visit the laboratory for posttesting after week 16 of the study. Posttesting included densitometry, strength testing, and blood and urine samples. The subjects randomized to a resistance-training group were asked to visit the laboratory twice per week on nonconsecutive days to participate in eccentric PRT for 16 weeks (32 training days).

Strength Evaluation

Strength testing was performed at pre-entry, baseline, and week 16 in all groups and approximately every 2 weeks in the treatment groups to maintain the appropriate target training intensity. Since eccentric 1-repetition maximum (1RM) loads are difficult to assess (very subjective) and are extremely large relative to the subject's concentric maximum strength, potentially increasing the risk for injury, muscle strength for the 6 exercises in this study were determined by the established concentric 1RM method (4). Furthermore, cross-training effects with eccentric resistance training that result in concentric strength gains have been reported (8, 11). The 1RM method was defined as the weight that can be lifted through a defined range of motion no more than once using proper form. Strength testing was preceded by 5 minutes of warm-up on a cycle ergometer. Subjects received detailed instructions on each exercise and performed each exercise several times at very low resistance to enhance familiarization and warm-up. The goal was to produce a 1RM within 3–5 efforts to reduce the effect of fatigue. To account for learning and familiarization with the strength-testing procedure, the pre-entry and baseline 1RM tests were separated by 5–7 days. The greatest 1RM value achieved between the 2 strength testing days was record-

TABLE 2. Eccentric resistance training protocol.*

Exercise	Low-intensity RT	High-intensity RT
Sessions/wk	2	2
Intensity† (%)	75	125
Sets	3	3
Repetitions	10	6
Training volume‡ (intensity × sets × repetitions)	22.5	22.5

* RT = resistance training.
† Percentage of concentric 1 repetition maximum.
‡ Calculated using the numbers in the table to demonstrate the similar training volumes between groups.

ed and represented the subject's maximum strength for the particular exercise tested.

Resistance-Training Intervention

The training intervention was conducted 2 times per week for 16 weeks (Table 2). No precedent exists for eccentric training; therefore, 2 training days per week were chosen for 2 reasons: (a) to allow appropriate rest between sessions, since eccentric contractions may generate extensive muscle damage with associated soreness; and (b) to provide feasibly for the managed training of 28 subjects, which resulted in the order of 37 hours of trainer-assisted exercise per week. Each training session lasted approximately 40 minutes and included 6 exercises: seated chess press, lat pull down, standing biceps curl, standing triceps extension, seated single-leg extension, and seated double leg curl. All exercises consisted of only eccentric muscle action with no concentric component. This was achieved by having the subject lower the weight against gravity (for 4 seconds) while the trainer lifted the weight through the concentric component of the exercise. The HRT group performed 3 sets of 6 repetitions at an intensity that was 125% of their concentric 1RM. The LRT group performed 3 sets of 10 repetitions at an intensity that was 75% of their concentric 1RM for each exercise (Table 2). The subjects in the LRT group performed 10 repetitions, as opposed to the 6 repetitions performed by the HRT group, to ensure that the training volume performed was the same for both groups.

Dual-Energy X-ray Absorptiometry

Bone mineral density (BMD) and bone mineral content (BMC) of the total body, proximal femur, and lumbar spine were determined by dual-energy x-ray absorptiometry (DXA; Hologic QDR-1500, software version 7.2). Additionally, lean mass and fat mass were measured from the whole-body scan. Scans were performed at baseline and week 16 for all subjects to provide total-body and regional values of bone, lean, and fat mass. The whole-body, proximal femur, and lumbar spine scans were performed with the subject positioned according to the manufacturer's specifications. The coefficient of variation for measures of bone mass in the laboratory is less than 1%.

Besides providing bone, lean, and fat mass measures of the total body, DXA also provides the ability to perform regional analyses. To corroborate the findings of our previous eccentric study in young women (8), mid-femur segment BMD, BMC, lean mass, and fat mass were determined from the whole-body scan as a measure of site-specificity. The mid-femur analysis was accomplished by shifting the whole-body scan into a function that allows the technician to determine the segmentalization of the image (8). The segment reported in this study was the middle 3 pixels measured between the middle of the knee joint and the top of the greater trochanter. Reproducibility of BMD values, assessed in 10 healthy volunteers, ranged from 0.5 to 1.0% for the segment determination.

Biochemical Markers

Blood and urine samples were collected in the morning prior to exercise and strength testing for analysis of osteocalcin, deoxypyridinoline, creatine kinase (CK), and creatinine. Approximately 6 ml of blood was drawn from an antecubital arm vein into a serum vacutainer tube with the subject at rest. The sample was allowed to clot for 15–30 minutes, immediately centrifuged for 10 minutes at 3,000*g,* pipetted into storage vials, and frozen at −80° C until analysis. Urine samples were taken from the second morning void. One and one-half millimeters of urine was pipetted into storage vials and frozen at −80° C until analysis. All baseline and week 16 samples were batched for analysis and run in duplicate. Samples were assayed by enzyme-linked immunoassay (ELISA) or enzyme immunoassay using the Dynex (Dynex Technologies, Inc., Chantilly, VA) Opsys MR plate reader. Values were calculated from a calibration curve fit (sigmoid) with a 4-parameter logistic equation using software provided by the manufacturer (Revelation QuickLink 4.04).

Urinary deoxypyridinoline, a marker of bone resorption, was measured using the Pyrilinks (R)-D ELISA assay kit (Metra Biosystems, Mountain View, CA). Serum osteocalcin, a marker of bone formation, was measured using an ELISA kit (Diagnostic Systems Laboratories, Inc., Webster, TX). The deoxypyridinoline intra- and interassay precision coefficient of variation was 4.7 and 5.2%, respectively, and values were corrected for differences in urine concentration by dividing by urinary creatinine analyzed using the Kodak Ectachem DT-60 II (Johnson & Johnson Clinical Diagnostics, Los Angeles, CA). The osteocalcin intra- and interassay precision coefficient of variation was 6.5 and 4.8%, respectively.

Serum CK was analyzed by standard spectrophotometric techniques (Ektachem DT II System, Johnson & Johnson, Rochester, NY), and these values were used to represent the magnitude of muscle damage associated with eccentric PRT.

Muscle Soreness

In addition to assessing muscle damage by serum CK levels, a perceived sensation of intensity of soreness scale was administered once per week. The scale ranged from no soreness to extreme soreness (17). Prior to exercise, subjects were asked to choose the appropriate indicator on the scale representing their peak soreness since the last training day.

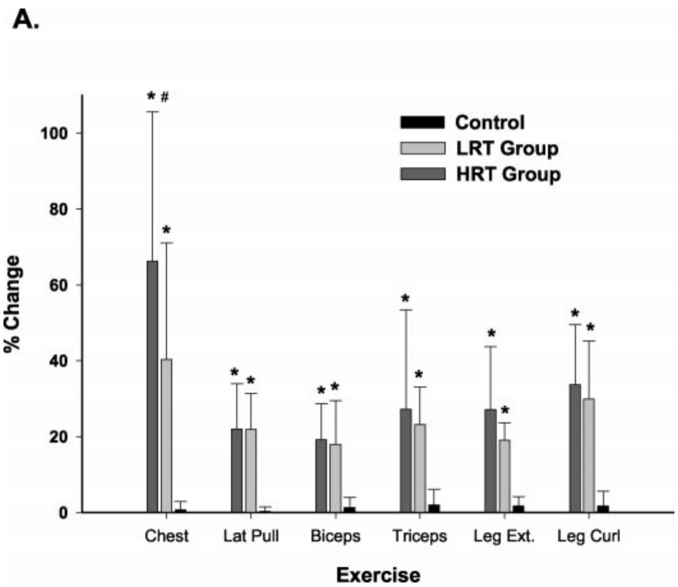
Menstrual, Physical Activity, and Health History Questionnaire

To assess the subjects' previous and current activity and health status, we asked them to complete a menstrual, physical activity, and health history questionnaire. The questionnaire included information about age at menarche, current menstrual history, current and past OC use, participation in sports and exercise, and family health

TABLE 3. Baseline and week 16 concentric strength (kg) by 1RM.*†

Exercise	Low-intensity RT (n = 14)		High-intensity RT (n = 14)		Control (n = 9)	
	Baseline	Week 16	Baseline	Week 16	Baseline	Week 16
Chest press	25.1 ± 7.5	39.9 ± 9.2‡	28.2 ± 10.1	37.0 ± 5.6‡	30.9 ± 5.1	31.2 ± 5.3
Lat pull	34.7 ± 6.9	42.0 ± 7.7‡	34.6 ± 6.8	41.9 ± 7.2‡	35.4 ± 5.8	35.5 ± 5.7
Biceps curl	20.5 ± 4.4	24.1 ± 4.3‡	20.7 ± 4.5	24.1 ± 3.9‡	21.5 ± 3.4	21.8 ± 3.3
Triceps extension	14.4 ± 4.4	18.0 ± 5.0‡	14.6 ± 2.9	17.9 ± 3.2‡	15.0 ± 3.7	15.4 ± 4.3
Leg extension	49.2 ± 10.1	61.2 ± 8.0‡	50.0 ± 7.9	59.4 ± 8.9‡	46.5 ± 7.5	47.2 ± 7.4
Leg curl	39.8 ± 7.4	52.6 ± 8.9‡	40.3 ± 7.2	51.7 ± 7.5‡	36.1 ± 7.8	36.7 ± 7.7

* RM = repetition maximum; RT = resistance training.
† Mean ± 1 *SD*.
‡ Significantly greater than baseline.



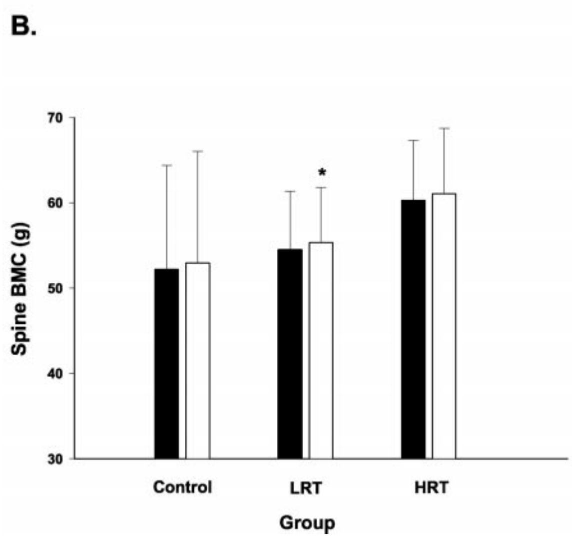


FIGURE 1. (a) Relative (%) change in strength from baseline to week 16. LRT represents the low-intensity progressive resistance-training group. HRT represents the high-intensity progressive resistance-training group. *Significant increase from baseline to week 16, $p < 0.05$. #Significantly greater than the LRT group, $p < 0.001$. (b) Spine bone mineral content (BMC) for baseline (filled bars) and week 16 (open bars) as measured by dual-energy x-ray absorptiometry (DXA). *Significantly greater than baseline values, $p < 0.05$.

history. Reported information was confirmed by oral interviews.

Statistical Analyses

Data were analyzed using the Statistical Package for Social Sciences software version 9.0 (SPSS Inc., Chicago, IL). A 2-way analysis of variance (ANOVA) with repeated measures was used to determine differences between and within groups for the dependent variables at baseline and week 16 measures. Additionally, a 2 × 5 (group × time) ANOVA with a repeated-measures design was used to test for significant main and interaction effects for the LRT and HRT groups. Tukey post hoc pairwise comparisons were performed in the case of significant F scores. A bidirectional α level of significance was set at $p \leq 0.05$ for all measures.

RESULTS

Strength

Baseline measures of 1RM concentric strength were similar in the LRT, HRT, and control groups (Table 3). Following 16 weeks of study intervention, the absolute and relative gains in 1RM concentric strength increased similarly in both the LRT and HRT groups, with exception of the chest press, which was almost twofold greater in the HRT group (8.7 ± 7.6 kg, mean ± *SD* vs. 14.8 ± 5.6 kg, respectively, $p < 0.001$). Additionally, the relative (%) gains in 1RM concentric strength in both groups were significantly different from the absence of change in strength in the control group (Figure 1a).

Body Composition

Following 16 weeks of study intervention, lean mass significantly increased in both the LRT (0.7 ± 0.6 kg, p = 0.002) and HRT (0.9 ± 0.9 kg, p = 0.006) groups, which were not significantly different between groups (F = 1.54, p = 0.228). No significant changes were demonstrated in the control group. Fat mass significantly increased in the LRT group (0.3 ± 0.6 kg, p = 0.05), while no significant change in fat mass was demonstrated in either the HRT or control group. These alterations in lean and fat mass resulted in significant total-body weight gain only in the LRT group (1.0 ± 0.8 kg, $p < 0.001$).

Bone Mass

At baseline, measures of BMC and BMD were similar in the LRT, HRT, and control groups (Table 4). Following 16 weeks of study intervention, no change in BMD was measured, with change occurring only in the BMC of the

TABLE 4. Baseline and week 16 BMC and BMD measures by DXA.*†

Bone mass	Low-intensity RT (n = 14)		High-intensity RT (n = 14)		Control (n = 9)	
	Baseline	Week 16	Baseline	Week 16	Baseline	Week 16
BMC (g)						
Spine	54.5 ± 6.9	55.4 ± 6.5‡	60.2 ± 7.0	61.1 ± 7.7	52.2 ± 12.2	52.9 ± 13.1
Proximal femur	31.1 ± 4.0	31.4 ± 4.3	31.6 ± 5.5	32.1 ± 5.5	25.9 ± 5.6	26.9 ± 5.8
Mid-femur	16.2 ± 2.2	16.3 ± 2.2	16.6 ± 2.6	16.7 ± 2.7	15.5 ± 3.5	15.6 ± 3.8
Total body	2,187 ± 172	2,192 ± 182	2,239 ± 339	2,252 ± 359	2,070 ± 483	2,094 ± 508
BMD ($g \cdot cm^{-2}$)						
Spine	1.03 ± 0.07	1.03 ± 0.07	1.07 ± 0.09	1.06 ± 0.09	0.98 ± 0.01	0.98 ± 0.01
Proximal femur	0.98 ± 0.10	0.97 ± 0.10	1.01 ± 0.12	1.01 ± 0.12	0.89 ± 0.01	0.90 ± 0.01
Mid-femur	1.44 ± 0.14	1.41 ± 0.15	1.46 ± 0.14	1.48 ± 0.17	1.41 ± 0.14	1.40 ± 0.17
Total body	1.06 ± 0.03	1.06 ± 0.04	1.09 ± 0.08	1.09 ± 0.09	1.03 ± 0.09	1.04 ± 0.08

* BMC = bone mineral content; BMD = bone mineral density; RT = resistance training; DXA = dual-energy x-ray absorptiometry.
† Mean ± 1 *SD*.
‡ Significantly greater than baseline.

TABLE 5. Baseline and week 16 deoxypyridinoline and osteocalcin values.*†

	Low-intensity RT (n = 14)	High-intensity RT (n = 14)	Control (n = 9)
Deoxypyridinoline ($nM \cdot mM^{-1}$)‡			
Baseline	4.7 ± 2.5	7.4 ± 8.8	6.9 ± 6.2
Week 16	4.6 ± 1.7	4.13 ± 4.7§	5.9 ± 6.6
Osteocalcin (ng·ml)			
Baseline	9.6 ± 5.3	15.1 ± 3.4	10.2 ± 4.5
Week 16	13.5 ± 5.4§	16.8 ± 4.4	9.8 ± 5.3

* RT = resistance training.
† Mean ± 1 *SD*.
‡ Deoxypyridinoline/urinary creatinine.
§ Significantly greater than baseline.

spine (0.855 ± 0.958 g, $F = 5.52$, $p = 0.005$) (Figure 1b) in the LRT group. No significant changes in BMC or BMD were found in the whole-body and proximal femur in any group (Table 4).

Bone Biochemical Markers

At baseline, measures of deoxypyridinoline crosslinks and osteocalcin were similar in the LRT, HRT, and control groups (Table 5). Following 16 weeks of study intervention, deoxypyridinoline in the LRT group did not significantly change. Furthermore, there were no significant differences between the 5 time points measured, although there was a trend toward increased deoxypyridinoline compared to baseline at week 8, after which they returned to baseline (Figure 2a).

Deoxypyridinoline levels in the HRT group followed a similar trend as in the LRT group, with the highest values peaking at week 8. However, in the HRT group, deoxypyridinoline decreased significantly from baseline (45 ± 89%, $p = 0.027$) and from week 4 (15.3 ± 20.7 $nM \cdot mM^{-1}$) to week 16 (4.13 ± 4.7 $nM \cdot mM^{-1}$, $p = 0.035$) (Figure 2a). Largely, the general trends of change in deoxypyridinoline were similar in both groups, with no significant difference observed between the change that occurred in the 2 groups ($F = 0.443$, $p = 0.643$); nor were interaction effects observed.

Osteocalcin significantly decreased (51 ± 48%, $p = 0.007$) in the LRT group during the first 8 weeks of the study (Figure 2b) and then increased significantly (165 ± 61%, $p < 0.001$) from weeks 8–16. Furthermore, following 16 weeks of intervention, osteocalcin increased significantly in the LRT group (31 ± 61%, $p = 0.037$) compared to baseline. Alterations in osteocalcin in the HRT group followed a similar trend as that found in the LRT group; however, the magnitude of change was greater. Osteocalcin decreased significantly (68 ± 27%, $p < 0.001$) in the HRT group during the first 8 weeks of the study intervention (Figure 2b). This relative decline in osteocalcin in the HRT group was significantly greater ($p = 0.030$) than the significant decline in osteocalcin in the LRT group from baseline to week 8. Following 8 weeks of intervention, osteocalcin demonstrated a significant (243 ± 68%, $p < 0.001$) increase from weeks 8–16, and this change was not significantly different between groups.

Serum CK

At baseline, measures of serum CK were similar in the LRT, HRT, and control groups. In the LRT group, serum CK significantly increased (60 ± 55%, $p = 0.025$) from baseline to week 4. The mean serum CK levels were maintained above baseline throughout the study in the LRT group but tended to decrease slightly after week 4 (Figure 3). Following 16 weeks of study intervention, the LRT group demonstrated a significant increase in serum CK of 47 ± 51% ($p = 0.025$). Similar to the LRT group, the HRT group showed a significant (35 ± 49%, $p = 0.021$) increase from baseline to week 4 of the intervention. In contrast to the LRT group, following study week

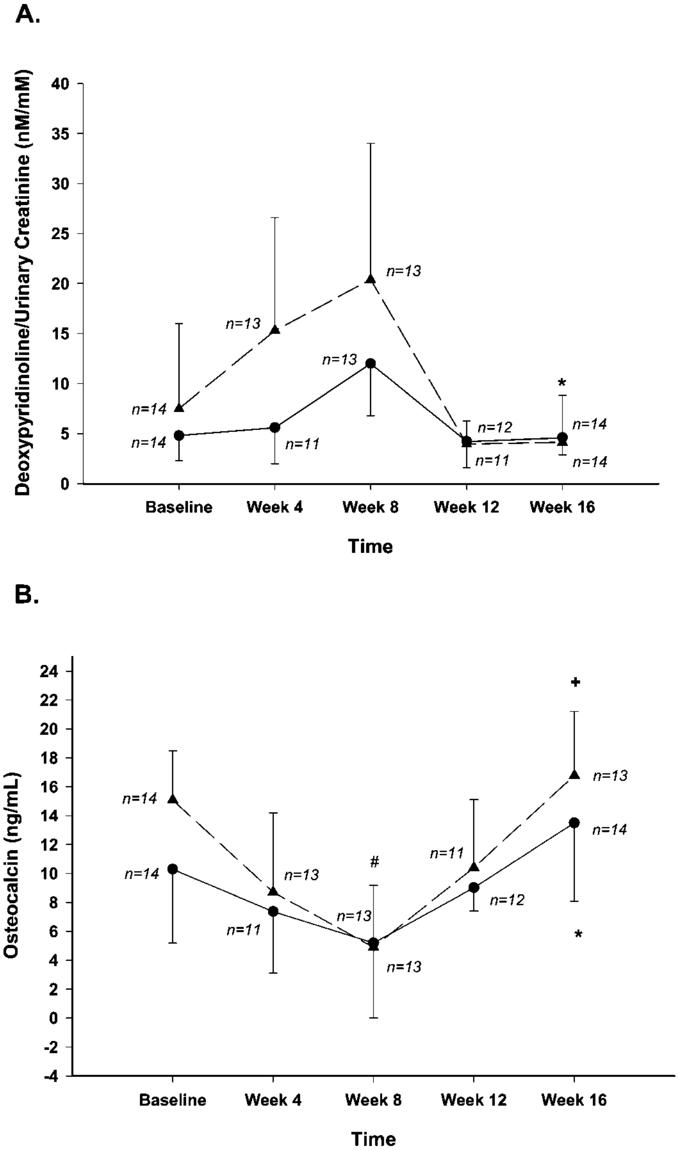


FIGURE 2. (a) Absolute change in deoxypyridinoline in the low-intensity progressive resistance-training (LRT) group (circles) and the high-intensity progressive resistance-training (HRT) group (triangles) from baseline to study week 16. *Significant decrease from baseline in the HRT group, $p < 0.05$. (b) Absolute change in osteocalcin in the LRT group (circles) and the HRT group (triangles) from baseline to study week 16. #Significant decrease from baseline in both the LRT and HRT groups. *Significant increase from baseline in the LRT group. +Significant increase from weeks 8–16 in both the LRT and HRT groups. $p < 0.05$ for all significant differences.

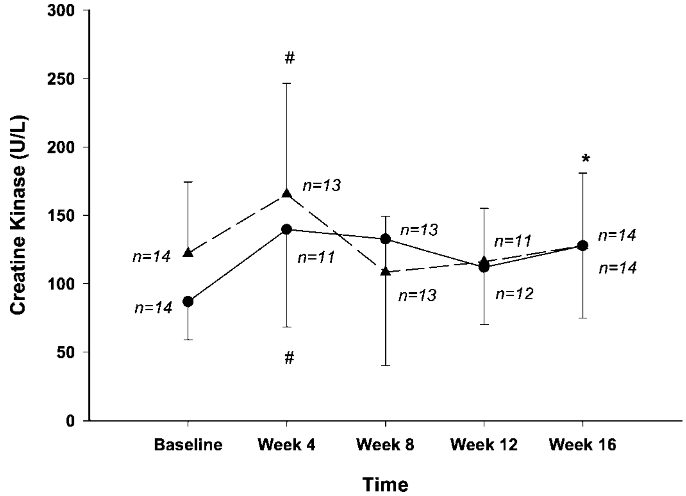


FIGURE 3. Absolute change in creatine kinase in the low-intensity progressive resistance-training (LRT) group (circles) and the high-intensity progressive resistance-training (HRT) group (triangles) from baseline to study week 16. #Significant increase from baseline in both the LRT and HRT groups. *Significant increase from baseline in the LRT group. $p < 0.05$ for all significant differences.

4, serum CK levels decreased significantly in the HRT group, returning to levels similar to baseline values (Figure 3). No significant changes in serum CK were observed in the control group.

Perceived Soreness

Subjects in the LRT group rated perceived soreness either slightly intense ($n = 9$) or intense ($n = 5$) the first 2 weeks of the study, after which their perceived soreness ratings were reduced to moderate ($n = 6$), mild ($n = 3$), or very mild ($n = 5$). Perceived soreness ratings in the HRT group were reported to be more intense, often receiving ratings of very intense ($n = 12$) or extremely intense ($n = 2$) on the soreness scale during the first 2 weeks of the study. However, 8 subjects and 6 subjects reported either feeling no soreness or very mild soreness, respectively, in the HRT group on subsequent assessments.

DISCUSSION

To our knowledge, this is the first study of young women to describe musculoskeletal adaptations to a whole-body eccentric PRT protocol using free weights and standard strength-training machines. Our findings provide important results of the effects of both LRT and HRT on skeletal muscle, bone mass, and bone biochemical markers and the efficacy of such training techniques. One of the most compelling findings of this study was that LRT (75% of a concentric 1RM) was as effective as HRT (125% of a concentric 1RM). This finding was unexpected, since it was hypothesized that the HRT group would be exposed to greater muscle loads and contraction-induced damage that would result in larger muscle and bone adaptations compared to the LRT group.

The most interesting findings of this study were the alterations in bone biochemical markers. First, there was a significant increase in osteocalcin (bone formation marker) with no change in deoxypyridinoline (bone resorption marker) in the LRT group. Elevated concentrations of osteocalcin in conjunction with decreased deoxypyridinoline would optimize the potential for enhanced bone mass by increasing formation and decreasing resorption, respectively. However, the absence of change in deoxypyridinoline combined with increased osteocalcin levels in the LRT group should ultimately result in augmented bone formation. Conversely, in the HRT group, deoxypyridinoline decreased significantly following 16 weeks of study intervention, while osteocalcin concentrations demonstrated a large nonsignificant increase. These findings suggest that the intensity of eccentric PRT influences the early patterns of bone adaptation to training. It is of interest that lower-intensity eccentric training influenced bone formation and

BMC, while high-intensity eccentric training was more influential in inducing adaptations in bone resorption. Further investigations of the influence of intensity on the patterns of bone adaptations to resistance training are warranted to allow us to differentiate whether different intensities of eccentric work influence bone through diverse mechanisms.

A limitation in interpreting the bone marker results is the inclusion of young women taking OCs. Since the hormonal milieu is an important regulator of the osteogenic response to loading, including women in the study who were using OCs may have confounded the findings. However, greater than 71% of the subjects in each training group were taking OCs, and when the data were analyzed controlling for OC status (ANCOVA), the results were similar to the reported findings. In any case, the measures of deoxypyridinoline and osteocalcin were highly variable, yet statistically significant, but may not necessarily connote physiologic significance.

The absence of significant alterations in bone mass in the HRT group was surprising, since we have previously demonstrated a significant 3.9% increase in mid-femur slice BMD in young women following 18 weeks of maximal eccentric training on an isokinetic dynamometer (8). The only significant bone mass adaptation in this study was demonstrated in the LRT group, with a 1.7% increase in spine BMC. Although statistically significant, this finding should be interpreted with caution, since there were no additional significant increases in BMC or BMD at any other site measured and, in fact, this may be only a spurious finding. While greater bone mass changes were hypothesized, particularly for the HRT group, these findings may have been limited by the short duration of the study intervention. The average turnover at a bone remodeling site is understood to be 6 months; however, bone turnover can occur in as few as 3 months or as long as 18 months (19). Most intervention training studies are conducted for less than 1 year because of logistic purposes, the associated high attrition rates, and evidence that longer-duration studies do not result in improved bone mass (5, 16). Although we cannot be certain that a longer training intervention would result in more impressive gains in bone mass, the alterations in bone biochemical markers in this study suggest the possibility. Thus, eccentric PRT may need to be continued longer than 16 weeks in order to yield significant increases in bone mass at multiple skeletal sites.

The resistance exercises chosen for this study may not have been optimal for stimulating an osteogenic response, particularly in the HRT group. Closed kinetic chain exercises, such as the squat, may have generated higher strain rates and magnitude than the exercises performed in this study, which could have maximized the osteogenic response to loading at very high intensities. However, since the high-intensity group was training at 125% of its concentric maximum, we believed that testing and training with extraordinary weight for an exercise that must be performed with exact form to avoid injury involved unwarranted risks in previously untrained individuals. Certainly, investigations using other closed kinetic chain exercises, such as the leg press, which may also load the proximal femur and spine without the potential risks or requiring the same skill to perform as the squat exercise, need to be conducted.

Analogous with our findings on bone mass, the modest and similar increases in lean body mass (0.7 ± 0.6 kg in the LRT group and 0.9 ± 0.9 kg in the HRT group) were unanticipated, since higher-force eccentric contractions are known to generate greater muscle damage than other contraction types and may therefore induce greater protein synthesis and, hence, greater muscle hypertrophy. Indeed, investigators have reported that eccentric contractions result in greater muscle hypertrophy than concentric contractions alone (9, 11, 18). Given the findings of this study, it appears that, in young women performing either LRT or HRT twice weekly, the stimulus for muscle hypertrophy under drastically different loads is similar. It may be possible that a low threshold response to eccentric training exists. However, it is important to point out the utility and safety of training at 75% of a concentric maximum that achieves the same benefits as training at supramaximal intensities.

Maximal voluntary muscle strength increased similarly in both the LRT and HRT groups, with the exception of the chest press, which was almost twofold greater in the HRT group. It is difficult to determine why eccentric training at an extremely high intensity did not result in greater increases in strength than training at a much lower intensity when it is well known that training concentrically with heavier loads results in greater strength gains than training with lighter loads (4). One possible explanation is that, regardless of training intensity, both eccentric-training groups performed the same training volume (intensity × sets × repetitions), suggesting that the total work performed was similar. However, a more likely explanation is the diminished influence of neural adaptations on strength gain associated with eccentric contraction. Since eccentric contraction requires less neural activation to generate the same force as a concentric contraction (8, 14), the neural adaptation and contribution to strength gain from eccentric training alone may be minimal.

Furthermore, the strength gains reported in this study may be underestimated, even though they are similar to the strength gains reported in previous resistance-training studies (7, 10). The concept of training specificity suggests greater gains in strength are measured when the movement pattern tested is the same as the movement pattern trained (i.e., eccentric strength testing of eccentric-trained muscles) (9). However, maximal eccentric strength is not effectively measured, since it requires timing the controlled lowering of a weight against gravity through a range of motion (a very subjective measurement), which may also jeopardize the safety of the trainee. Consequently, for practicality and safety, we elected to determine strength by the established 1RM method. The safety risk for both the trainee and trainer is reduced when lifting and lowering submaximal loads. Therefore, our findings of 20–40% similar gains in strength between the 2 groups suggest that training at 75% of a concentric 1RM is advantageous to training at 125% of a concentric 1RM.

We expected the HRT group to experience greater muscle damage than the LRT group, as determined by serum CK and perceived muscle soreness. While we recognize that the significant alterations in CK reported in this study are very modest, other investigators reporting elevated serum CK as a result of eccentric exercise have designed studies that maximally load a single muscle group for a long duration, generating much more extensive muscle damage than the short-duration multiple-muscle group exercises performed in this study. Typically, the higher the tension generated in a muscle by eccentric contraction, the greater the induced muscle damage (1). Although muscle tension was probably extremely high when the 6 repetitions were performed by the HRT group, the additional 4 repetitions per set performed by the LRT group might have provided enough low-tension–induced, repetitive muscle damage to produce the sustained elevated serum CK levels.

Alternatively, more important than the intensity of training may be the greater number of eccentric contractions performed in the LRT group. The findings from the perceived soreness scale used in this study corroborate the CK results of sustained albeit very modest muscle damage in the LRT group. Subjects in the LRT group rated perceived soreness either slightly intense or intense the first 2 weeks of the intervention, after which their perceived soreness ratings were reduced to moderate, mild, or very mild. However, at no time did any subject in the LRT group report the absence of soreness.

It is possible that the similar findings between groups in this study were the result of the high-intensity–trained subjects not being able to effectively lower the weight against gravity, which would diminish the tension generated by the active muscles. We chose the intensity of 125% of a concentric 1RM because intensities greater than 130–140% result in a resistance that cannot be controlled against gravity (12). Therefore, if the external force exceeded the subject's ability to control the force, then training at 125% intensity may have been ineffective. However, all subjects were asked to lower the weight on a 4-second count, demonstrating effective control of the load. Although we could not directly assess the in vivo muscle force production, we are confident that the subjects in the HRT group were performing supramaximal eccentric contractions in a controlled manner.

In conclusion, the findings of this study suggest that whole-body eccentric PRT results in sizable gains in muscle strength with modest increases in lean body mass, regardless of training intensity. Furthermore, the 20–40% gains in concentric strength following 16 weeks of eccentric PRT in young women were similar to the increases reported by other investigators using standard training protocols. However, the absence of a standard training group that included both concentric and eccentric movements limits the conclusions that can be drawn from these findings. The alterations in bone metabolic markers suggest that the intensity of eccentric PRT influences the early patterns of bone adaptation to training and that a longer training intervention, indeed, may result in enhanced bone mass. Lastly, load magnitude may not be the greatest contributor to bone adaptation from eccentric PRT, and it appears that submaximal intensities may be optimal for musculoskeletal adaptations. Further studies are needed to corroborate the findings of this investigation.

Practical Applications

Resistance training can augment muscle mass and strength and improve bone health in women. This is of particular importance in young women, since peak bone mass is generally not achieved until the age of 30 years, and resistance training during this time may enhance peak BMD. Because training exclusively with eccentric contractions requires the assistance of a trainer or partner or the use of specialized equipment, eccentric training has practical limitations. While advantages of eccentric training have been proposed, the findings of this study demonstrate that eccentric training at 75% of a concentric 1RM results in significant musculoskeletal adaptations. Additionally, this training intensity allows the trainee to physically perform the concentric movement without assistance. Therefore, the trainee would be able to concentrically move the resistance against the force of gravity and eccentrically control the resistance, returning to the starting position. This would be advantageous compared to training with eccentric loads that exceed a concentric 1RM, since that would require a trainer or partner or the use of special equipment and could jeopardize the safety of those involved. However, if submaximal eccentric training loads are performed in combination with concentric loads, as occurs with standard resistance training, the trainee should concentrate on the eccentric component by performing slow (4 seconds) controlled movements.

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